Constant-factor sharp dimension threshold for perfect matchings in regular Cartesian products

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Editorial note. The Cartesian edge-connectivity formula and Tutte-Berge formula are explicit permitted classical inputs. The quantitative threshold and the regular obstruction construction remain proof obligations.

Editorial note. The statement uses t>2 ln M, as printed in source103, rather than the weak inequality used in the abstract.

Editorial note. The phrase maximal matching in the last sentence of the auxiliary proof365 refers to a maximum matching: the displayed Tutte-Berge deficiency is defined for a maximum matching in source239. The author text is retained.

Editorial note. The lower-bound construction is phrased for sufficiently large M in source381. It establishes constant-factor asymptotic sharpness; the editorial title does not add a claim at small M.

Editorial note. The source disclosure at1686-1687 is retained: AI tools were used for polishing, while the authors state that all proof ideas were entirely human-generated.

Editorial scope. One deterministic two-sided dimension theorem, with its complete regular-graph lemma, lower-bound construction and native diagram. Random-process hitting times and their proof lemmas are outside this task.

Allowed standard prerequisites (editorial).

- Tutte-Berge formula. The matching deficiency equals max over U of odd(G minus U) minus |U|.. Source237-243, citing Berge1962; the exact formula is printed as allowed context.. Translate bounds on odd components, and the explicit constructed obstruction, into matching existence/nonexistence.
- Classical Cartesian-product edge-connectivity formula. For connected regular factors of sizes M_i and total degree d, lambda(product)=min(d,min_i lambda(H_i) product_(j!=i) M_j).. Source318-335, citing Hammack, Imrich and Klavzar, Handbook of Product Graphs (2011), Chapter5.. This exact classical structural formula is explicitly permitted; its proof is not part of the counted task.
- Elementary finite counting and real inequalities. Parity of a cut, exponential-versus-linear estimates, and the number/order of Cartesian-product components.

Given two graphs $H_1 = (V_1, E_1)$ and $H_2 = (V_2, E_2)$, the *Cartesian product* $H = H_1 \square H_2$ is the graph whose vertex set is $V_1 \times V_2$, and for $u_1, v_1 \in V_1$ and $u_2, v_2 \in V_2$, we have that $\{(u_1, u_2), (v_1, v_2)\} \in E(H)$ either if $u_1 = v_1$ and $\{u_2, v_2\} \in E(H_2)$ or if $u_2 = v_2$ and $\{u_1, v_1\} \in E(H_1)$. More generally, given t graphs, $H_i, \ldots, H_t$, their Cartesian product $G = \square_{i=1}^{t} H_i$ is the graph with the vertex set

$$V \coloneqq \{v = (v_1, \ldots, v_t) \colon v_i \in V(H_i) \text{ for all } i \in [t]\},$$

and the edge set

$$\left\{ uv \colon \begin{array}{c} \text{there is some } i \in [t] \text{ such that } \{u_i, v_i\} \in E(H_i) \\ \text{and } u_j = v_j \text{ for all } j \neq i \end{array} \right\}.$$

A *perfect matching* in a graph G is a collection of vertex disjoint edges which cover the vertex set of G. A matching is *nearly-perfect* if it leaves exactly one vertex unmatched. In this paper, we study perfect matchings of high-dimensional product graphs, both in the deterministic and in the random setup.

We begin with the deterministic part. Observe that if one base graph, say H_j, has a perfect matching, then $G = \square_{i=1}^{t} H_i$ has a perfect matching: for each fixed choice of the other coordinates we may take the corresponding copy of that matching in the j-th coordinate.

For a graph Γ and a set $A \subseteq V(\Gamma)$, write $E_\Gamma(A)$ for the set of edges of Γ with both endpoints in A, and set $e_\Gamma(A) \coloneqq |E_\Gamma(A)|$. For disjoint sets $A, B \subseteq V(\Gamma)$, write $E_\Gamma(A, B)$ for the set of edges of Γ between A and B, and set $e_\Gamma(A, B) \coloneqq |E_\Gamma(A, B)|$. We abbreviate $A^c = V(\Gamma) \setminus A$ when the graph in question is clear. Furthermore, we denote by $\lambda(\Gamma)$ the minimum number of edges whose removal disconnects Γ and call this the edge-connectivity of Γ. The deficit of Γ, abbreviated by $\mathrm{def}(\Gamma)$, is the number of vertices left unmatched by a maximum matching. Finally, we write $\mathrm{odd}(\Gamma)$ for the number of odd-order components in a graph Γ.

The Tutte-Berge formula The Tutte-Berge formula [1] is our first tool in order to treat the existence of perfect matchings. It implies that the number of unmatched vertices in a maximum matching in a graph Γ is equal to

$$\mathrm{def}(\Gamma) = \max_{U \subseteq V(\Gamma)} \{\mathrm{odd}\,(\Gamma[V(\Gamma) \setminus U]) - |U|\}, \tag{1}$$

where we recall that $\mathrm{odd}\,(\Gamma[V(\Gamma) \setminus U])$ is the number of connected components with an odd number of vertices in $\Gamma[V(\Gamma) \setminus U]$. In particular, a graph Γ has a perfect matching if and only if for every subset $U \subseteq V(\Gamma)$, the subgraph $\Gamma[V(\Gamma) \setminus U]$ has at most $|U|$ connected components with an odd number of vertices. Note here that if Γ has isolated vertices, then choosing $U = \varnothing$ witnesses that there is no perfect matching in Γ.

Lemma 0.1. *Let $G = \square_{i=1}^{t} H_i$, where every H_i is connected and d_i-regular. Let $M_i = |V(H_i)|$ and $d = \sum_{i=1}^{t} d_i$. Then*

$$\lambda(G) = \min \left\{ d,\ \min_{i \in [t]} \lambda(H_i) \prod_{j \neq i} M_j \right\}.$$

Proof. The displayed identity is the standard edge-connectivity formula for Cartesian products; see [2, Chapter 5]. □

Theorem 1. *Let $M \geq 2$ and let $t > 2\ln M$ be integers. For every $i \in [t]$, let H_i be a connected d_i-regular graph with $1 < |V(H_i)| \leq M$, and write $G = \square_{i=1}^{t} H_i$. Then G has a (nearly-)perfect matching.*

Furthermore, there is a sequence of such graphs $(H_i)_{i\in[t]}$ such that for $t < \ln M/3$ the product $G = \square_{i=1}^{t} H_i$ does not have a perfect matching.

Lemma 0.2. *Let $d \geq 1$ be an integer and let Γ be a connected d-regular graph. Suppose that $\lambda(\Gamma) \geq d-1$. Then Γ has a perfect matching when $|V(\Gamma)|$ is even and a nearly-perfect matching when $|V(\Gamma)|$ is odd.*

Proof. For $U \subseteq V(\Gamma)$, let $Q_1, \ldots, Q_q$ be the odd components of $\Gamma - U$ such that $q = \mathrm{odd}(\Gamma - U)$. If $U = \varnothing$, then $q = 0$ when $|V(\Gamma)|$ is even and $q = 1$ when it is odd. Suppose now that $U \neq \varnothing$. Every Q_i is a proper non-empty set, so by the assumption that $\lambda(\Gamma) \geq d-1$,

$$e_\Gamma(Q_i, Q_i^c) \geq d - 1.$$

Since $|Q_i|$ is odd and Γ is d-regular (implying that $d|Q_i| = 2e_\Gamma(Q_i) + e_\Gamma(Q_i, Q_i^c)$), we have

$$e_\Gamma(Q_i, Q_i^c) \equiv d|Q_i| \equiv d \pmod 2.$$

The cut has the same parity as d, whereas $d-1$ has the opposite parity; hence

$$e_\Gamma(Q_i, U) = e_\Gamma(Q_i, Q_i^c) \geq d.$$

Summing over the odd components yields

$$qd \leq \sum_{i=1}^{q} e_\Gamma(Q_i, U) \leq d|U|,$$

and therefore $q \leq |U|$. The Tutte–Berge formula (1) implies that all vertices are covered by a maximal matching if $|V(\Gamma)|$ is even and all but at most one vertex are covered by a maximal matching if $|V(\Gamma)|$ is odd. □

Proof of Theorem 1. Let $G = \square_{i=1}^{t} H_i$. If a base graph H_i has a perfect matching M_i, then for each fixed choice of all coordinates other than the i-th, take the corresponding copy of M_i. The resulting matchings are vertex-disjoint and cover $V(G)$.

Otherwise, any base graph H_i has at least three vertices. Thus, by Theorem 0.1 we obtain that

$$\lambda(G) = \min\left\{ d,\ \min_{i\in[t]} \lambda(H_i) \prod_{j\neq i} |V(H_j)| \right\} \geq \min\{d, 3^{t-1}\}. \tag{2}$$

Using that $d \leq t(M-1)$ and $t > 2\ln M$ we obtain that (2) is at least $d-1$. Applying Theorem 0.2, we obtain a (nearly-) perfect matching. This concludes the first part of the theorem.

For the second part, we provide an explicit construction where for $t < \ln M/3$ there is no perfect matching. Indeed, let M be sufficiently large, and let m be

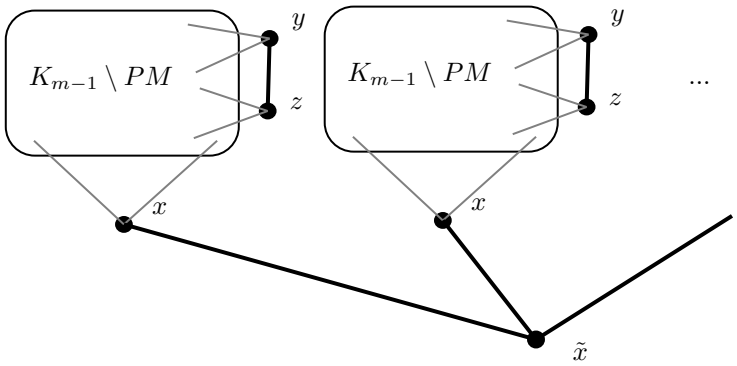


Figure 1: The construction of H.

the largest odd integer with $(m+1)^2 \leq M$. Construct a graph H' as follows. Consider K_{m-1} and three distinct vertices x, y, and z. Remove one perfect matching from K_{m-1}, draw the edge yz, and draw all edges between x, y, z and K_{m-1}. Thus the vertex x has degree $m-1$, and every other vertex in H' has degree m. We then construct H by taking m copies of H' and a distinct vertex $\tilde{x}$, and drawing the edges between $\tilde{x}$ and the copy of x in each copy of H', see Figure 1. Thus H is an m-regular graph on $(m+1)^2$ vertices. Moreover, since m is odd, $H \setminus \{\tilde{x}\}$ has m odd components, and thus H has no perfect matching by Tutte's Theorem. Finally, construct G by taking the Cartesian product of H with $(t-1)$ triangles K_3. The base graphs all have size at most M, are connected and regular. Consider the set $U = \{\tilde{x}\} \times V(\square_{i=1}^{t-1} K_3)$. $G - U$ has exactly m odd components, so if $|U| = 3^{t-1} < m \leq \sqrt{M}$, G has no perfect matching. In particular, any $t < \ln M/3$ suffices. □

Statement about the use of AI The authors acknowledge the use of AI tools for polishing this paper, and note that all of the proof ideas were entirely human-generated.

References

[1] Claude Berge. *The theory of graphs and its applications.* London: Methuen & Co. Ldt.; New York: John Wiley & Sons. Inc. X, 247 p. (1962)., 1962.

[2] Richard Hammack, Wilfried Imrich, and Sandi Klavžar. *Handbook of product graphs.* Discrete Mathematics and its Applications (Boca Raton). CRC Press, Boca Raton, FL, second edition, 2011. With a foreword by Peter Winkler.